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Aryabhatta Knowledge University, Patna

B.Tech 5th Semester Examination, 2020

Time: 03 Hours

Code: 101507 ***Subject: Transportation Engineering*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option from the following (any seven): **[14]**

(a) Minimum of 8 m width and 2 m wide shoulders on the following roads are recommended
(i) EH and NH
(ii) NH and SH
(iii) EH, NH and SH
(iv) MDR

(b) Which of the following statements is correct? According to the principle of road alignment
(i) it should be straight.
(ii) its gradient should be easy.
(iii) it should aim avoiding deep cuttings and fillings.
(iv) All of the above

(c) In the final survey, the centreline stakes are driven at ___ interval in hilly terrain.
(i) 10 m
(ii) 20 m
(iii) 30 m
(iv) 50 m

(d) The portion of the roadway between outer edges of carriageway and drains in case of cuttings is known as
(i) kerb
(ii) shoulder
(iii) formation width
(iv) right of way

(e) Maximum camber recommended by IRC is
(i) 1.7%
(ii) 2%
(iii) 3%
(iv) 4%

(f) For highways, width of the formation in plain area is
(i) 18 m
(ii) 15 m
(iii) 12 m
(iv) 9 m

(g) In steep terrain, ruling gradient is limited to
(i) 1 in 25
(ii) 1 in 20
(iii) 1 in 17
(iv) 1 in 12.5

(h) For finding stopping distance of a vehicle, the height of line of sight of driver and height of line of obstacle on road are taken as ___ respectively.
(i) 1.2 m and 0.15 m
(ii) 1.5 m and 0.5 m
(iii) 1.2 m and 0.5 m
(iv) 1.5 m and 0.15 m

(i) When a vehicle traces a horizontal curve, it is subjected to centrifugal force in ___ direction.
(i) inward
(ii) outward
(iii) forward
(iv) backward

(j) For designing highways, IRC recommends the coefficient of friction as
(i) 0.15
(ii) 0.18
(iii) 0.21
(iv) 0.24

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Q.2 Solve both questions : **[14]**

(a) Outline and discuss the scope of the highway engineering study.

(b) Briefly discuss the historical development of road construction.

Q.3 Solve this question : **[14]**

There are five alternative proposals of road plans for a backward district. The details are given below. Justify with reasons, which proposal is the best, assuming utility unit of 0.5, 1.0, 2, 4 and 8 for the five population ranges and 1.0 and 5 per 1000 t of agricultural and industrial products served:

Proposal	Total length (km)	Number of towns and villages served with population range					Productivity (in Thousand tonnes)	
		<2000	2001-5000	5001-10000	10001-20000	>20000	Agricultural	Industrial
P	500	100	150	20	40	3	150	20
Q	600	200	250	68	28	3	220	25
R	700	270	350	82	36	4	300	35
S	800	280	410	91	41	4	400	42
T	900	290	430	96	44	4	430	45

Q.4 Solve both questions : **[14]**

(a) What are the various requirements of an ideal highway alignment? Explain briefly.

(b) Discuss the general principles in the re-alignment of a highway and explain how the work is carried out.

Q.5 Solve this question : **[14]**

A national highway passing through a rolling terrain in heavy rainfall area has a horizontal curve of radius 500 m. Design the length of transition curve, assuming suitable data.

Q.6 Solve both questions : **[14]**

(a) State the factor on which the overtaking sight depends. Explain briefly.

(b) State the objects of widening pavements on horizontal curve. What are the factors on which the design of widening depends? Explain.

Q.7 Solve this question : **[14]**

The 15-minute traffic counts on crossroads 1 and 2 during peak hour are observed as 178 and 142 vehicles per lane respectively, approaching the intersection in the direction of heavier traffic flow. If the amber time required are 3 and 2 seconds respectively for load based approach speed, design the signal timings by trial cycle method. Assume an average time headway of 2.5 seconds during green phase.

Q.8 Solve both questions : **[14]**

(a) Explain briefly the modified Hubbard Field method of bituminous mix design.

(b) Explain the desirable properties of aggregate to be used in different types of pavement construction.

Q.9 Solve both questions : **[14]**

(a) Write down the construction steps for water bound macadam road.

(b) With the aid of neat sketches, show the different types of joint and their positions.

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